Guest Editorial: **A Simplified Diabetic Retinopathy Scale**

I congratulate the authors of this monumental effort to develop an international classification of diabetic retinopathy severity and diabetic macular edema.[1] To reach consensus on this difficult topic among 30 participants from 16 countries is indeed an accomplishment. As we observe a dramatic increase in the number of persons afflicted with diabetes worldwide, I hope this work will lead to a significant step in our efforts to reduce diabetes-related blindness. Approximately 120 million people are estimated to have diabetes throughout the world. Currently, 20% of the population 65 or older have diabetes in the United States. Whereas 17 million Americans have diabetes now, it is estimated that, by the year 2025, 25 million will have diabetes. Caring for these patients will be a big task. The development of a simplified classification is an important initial step in evaluating and caring for our patients with diabetes.

An important purpose of this new classification is to provide common language to facilitate communication between general ophthalmologists and retinal specialists around the world. Communication between the medical doctors and the ophthalmologists is essential, because glucose control, blood pressure control, and management of dyslipidemia play important roles in preventing the development of or retarding the progression of diabetic retinopathy. Data from clinical trials show that intensive glucose control may prevent the progression of diabetic retinopathy by as much as 50% to 75%, whereas blood pressure control may reduce the rate of moderate vision loss by as much as 47%.[2,3] Observational data suggest that dyslipidemia may result in vision loss as well as increase the risk of developing proliferative diabetic retinopathy.[4,5] The importance of medical management in patients with diabetic retinopathy cannot be emphasized enough, and enhanced communication between ophthalmologists, patients, and all those who take care of them can be facilitated with the use of this simplified classification.

Is this a new system of classifying diabetic retinopathy? Obviously, the development of the current scales heavily depends on the analyses and knowledge gained from a number of landmark studies of diabetic retinopathy, including the Early Treatment Diabetic Retinopathy Study (ETDRS), the Wisconsin Epidemiologic Study of Diabetic Retinopathy, and others.[6–8] These studies have provided the basic knowledge for understanding the progression rate of diabetic retinopathy, the associated risk factors, and the outcomes of treatment. The ETDRS classification system is the gold standard used in many clinical studies of diabetic retinopathy, both controlled clinical trials and epidemiologic studies. This system is based on the grading of fundus photographs of seven stereoscopic fields. Each lesion is evaluated in each of the nine subfields, and the severity of the retinopathy is summarized after complex assessments of the lesions. The ETDRS classification is indeed an important tool for clinical research, but it is not practical for clinical use. The new international clinical system is not intended to replace the ETDRS scale in clinical research but to put in practice the clinical guidance provided by these important clinical studies.

This new classification is intended to be used by the international community of ophthalmologists and their assistants (e.g., nurse practitioners and other medical personnel who in many cases will not have access to biomicroscopic methods of detecting diabetic retinopathy). The system needs to be flexible enough to allow for recognition of important ocular lesions with simple direct ophthalmoscopy to classify retinopathy severity. This system should be clinically meaningful and easily adapted for clinical practice. For most of the stages of severity of retinopathy there was considerable agreement among the authors, indicating a certain amount of ease in recognizing these stages.

I commend the authors for the inclusion of the severe nonproliferative stage in the scale, despite the disagreement among the authors, because this is a critically important clinical stage in the classification. Data from the ETDRS showed that early treatment beginning with laser photocoagulation for patients with type 2 diabetes and the severe nonproliferative stage reduced the rate of vision loss by as much as 50%.[9] Venous abnormalities and intraretinal microvascular abnormalities may be more difficult to recognize, thus making this stage somewhat difficult to diagnose. The authors have included a sample of these lesions on fundus photographs to help those who are not familiar with them. In addition, it appears that microaneurysms and blot hemorrhages, the most readily recognizable lesions, may be the presenting signs of this severe stage in more than 50% of cases.[6] This may help to decrease the likelihood of missing this important stage in which treatment may be initiated.

The difficulties in the recognition of some of the ocular lesions, such as the venous abnormalities and intraretinal microvascular abnormalities, indicate the importance of education of general ophthalmologists, retinal

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specialists in particular, and the medical physicians. The new grading system needs to be validated through clinical studies in which the clinical assessments of the treating ophthalmologists are compared with the detailed fundus photographic grading of seven standard stereoscopic fields. One can test the sensitivity and specificity of such a system. A relatively easy way of testing this classification would be the inclusion of an ophthalmologist's clinical grading in a study that already uses the seven standard stereoscopic fundus photographs. Ideally, the clinical assessments from both the general ophthalmologists and the retinal specialists on the same patient population should be included. The photographic grading would be compared with the simplified clinical grading. A goal of a "validation" study would be to evaluate the loss of information when the simpler scheme is used. Such a study may appear simple but will require significant resources to conduct.

Are there weaknesses to the international classification? A potential one is the diagnosis of diabetic macular edema. Although less severe than high-risk proliferative disease, macular edema is more prevalent. The data regarding edema detection from previous studies depend on stereoscopic viewing, either clinically or photographically of the macula, and the use of fluorescein angiography. Because this system has to be functional in conditions where such sophisticated technology may not be readily available, the international classification relies heavily on the recognition of retinal hard exudates. The detection of retinal thickening without the stereoscopic technology either clinically or photographically is challenging. The number of cases of clinically significant macular edema that would be considered for treatment should be carefully evaluated in the study mentioned above to test the sensitivity and specificity of the proposed system for diabetic macular edema. A reduction in both sensitivity and specificity may have to be sacrificed for the ease of use of this system throughout the world.

In addition to obtaining the sensitivity and specificity of this international classification in clinical studies, it will be important to measure the medical impact of the use of this system. Will this simple grading lead ultimately to a greater number of patients being treated appropriately in a timely fashion? Education about the indications for treatment should be an integral part of the introduction of the proposed international classification. Ultimately, it will be important to measure whether such a clinical system will lead to better care and a decrease in blindness from diabetic retinopathy.

The development of this international classification of diabetic retinopathy for both retinopathy severity and diabetic macular edema is a very important step towards better care of patients with diabetes throughout the world. As the number of people affected with diabetes grows exponentially, this system should help to facilitate communication among the medical team. Hopefully, through improved communication, the importance of the medical management of diabetes and its complications, including diabetic retinopathy, will be recognized and embraced by both physicians and patients. It is up to us, the ophthalmologists taking care of diabetic patients, to put this international classification to test: to evaluate its validity and its ease of use. We need to combine our efforts as researchers and physicians to take the classification to the next level and to participate in the education of our fellow ophthalmologists and medical colleagues about this classification. The challenge of delivering timely eye care to *all* persons with diabetic retinopathy throughout the world remains.

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